Can Regressive Green Subsidies Be Justified by Abatement and Efficiency?

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Abstract

Green subsidies for households often accrue disproportionately to those with high incomes. Can this regressivity be justified on environmental grounds? The case rests on two premises: that richer households abate more, and that their abatement costs the government less per tonne. We quantify subsidy-induced CO_2 abatement and its costs across the income distribution for the Dutch heat-pump subsidy (ISDE), linking the full daily record of applications to administrative income and smart-meter energy data for the universe of Dutch homeowners. We estimate the adoption response to an unanticipated increase in the subsidy rate using a regression discontinuity in time, and per-installation abatement using a staggered difference-in-differences. Combining these estimates, we find that CO_2 abatement rises steeply with income, but inframarginal spending rises in step with it, leaving the public cost per tonne abated nearly flat. Efficiency therefore provides no argument for the regressivity; only the level of abatement does. A Marginal Value of Public Funds calculation reveals that welfare per euro of public spending is highest among low-income households, driven primarily by the higher value of a euro reaching them.

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